

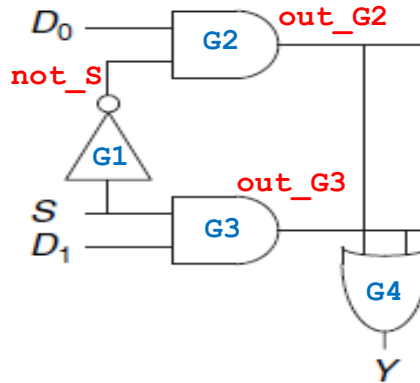
Lab no 02: Simulate 2-to-1 Multiplexer

The purpose of this Lab is to learn to: Simulate a 2-to-1 multiplexer on ModelSim and verify the multiplexer function using testbench. In this lab, you will build the multiplexer using logic gates (refer to Lab 01).

Parts: -

Simulate 2-1 multiplexer using logic gates.

Simulate 2-1 multiplexer using logic gates (Lab01).



The multiplexer chooses between the two data inputs based on the select:

if $S = 0$, $Y = D_0$, and if $S = 1$, $Y = D_1$.

Verilog code for multiplexer 2-to-1

```
// ----- Mux 2-1 -----//
module mux_2_1(S,D0,D1,Y);
input S, D0, D1;
output Y;
wire not_S, out_G2, out_G3;
notGate G1(S, not_S);
andGate G2(D0, not_S, out_G2);
andGate G3(D1, S, out_G3);
orGate G4(out_G2, out_G3, Y);
endmodule

// ----- AND Gate -----//
module andGate (a,b,c);
input a, b;
output c;
assign c = a & b;
endmodule

// ----- Or Gate -----//
```

```
module orGate (a,b,c);  
input a, b;  
output c;  
assign c = a | b;  
endmodule  
  
// ----- Not Gate -----//  
module notGate (a,c);  
input a;  
output c;  
assign c = ~a;  
endmodule
```

Testbench of the multiplexer 2-to-1

```
module mux_tb;  
    reg s, d0, d1;  
    wire y;  
    mux_2_1 mux_dut(s,d0,d1,y);  
initial  
    begin  
        s= 0; d0 = 0; d1 =0;  
        #10 s=0; d0=0; d1=1; // select d0=0  
        #10 s=0; d0=1; d1=0; // select d0=1  
        #10 s=1; d0=0; d1=1; // select d1=1  
        #10 s=1; d0=1; d1=0; // select d1=0  
    end  
endmodule
```